07 . Use K-Medoids clustering to segment wholesale customers based on their annual spending on different product categories. Identify the optimal number of clusters to effectively group customers with similar purchasing behaviours.

In [3]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

In [4]:
```
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.metrics import *
from kmedoids import KMedoids
```

In [6]:
```
df = pd.read_csv('Wholesale_Customers.csv')
print("Dataset Shape :", df.shape)
df.sample(5)
```

```
Dataset Shape : (440, 9)
```

Out[6]:

	Channel	Region	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicassen	Cluster
265	1	1	5909	23527	13699	10155	830	3636	1
231	1	1	2344	10678	3828	1439	1566	490	1
268	2	1	11908	8053	19847	1069	6374	698	1
227	1	1	4042	2204	1563	2286	263	689	1
322	1	2	15881	713	3315	3703	1470	229	1

In [7]:
```
if df.isnull().sum().sum() > 0 :
    print("Missing Values Present")
    print(df.isnull().sum())
else:
    print("No Missing Values Found")
```

```
No Missing Values Found
```

In [8]:
```
df.columns
```

Out[8]:
```
Index(['Channel', 'Region', 'Fresh', 'Milk', 'Grocery', 'Frozen',
       'Detergents_Paper', 'Delicassen', 'Cluster'],
      dtype='object')
```

In [11…
```
features = ['Fresh', 'Milk', 'Grocery', 'Frozen', 'Detergents_Paper', 'De
```

In [12…
```
x = df[features]
```

In [13…
```
scaler = StandardScaler()
x_scale = scaler.fit_transform(x)
```

In [14…

```
scores = []

for k in range(2,11) :
    model = KMedoids(n_clusters=k, random_state=42, metric='euclidean', m

    labels = model.fit_predict(x_scale)

    score = silhouette_score(x_scale, labels)

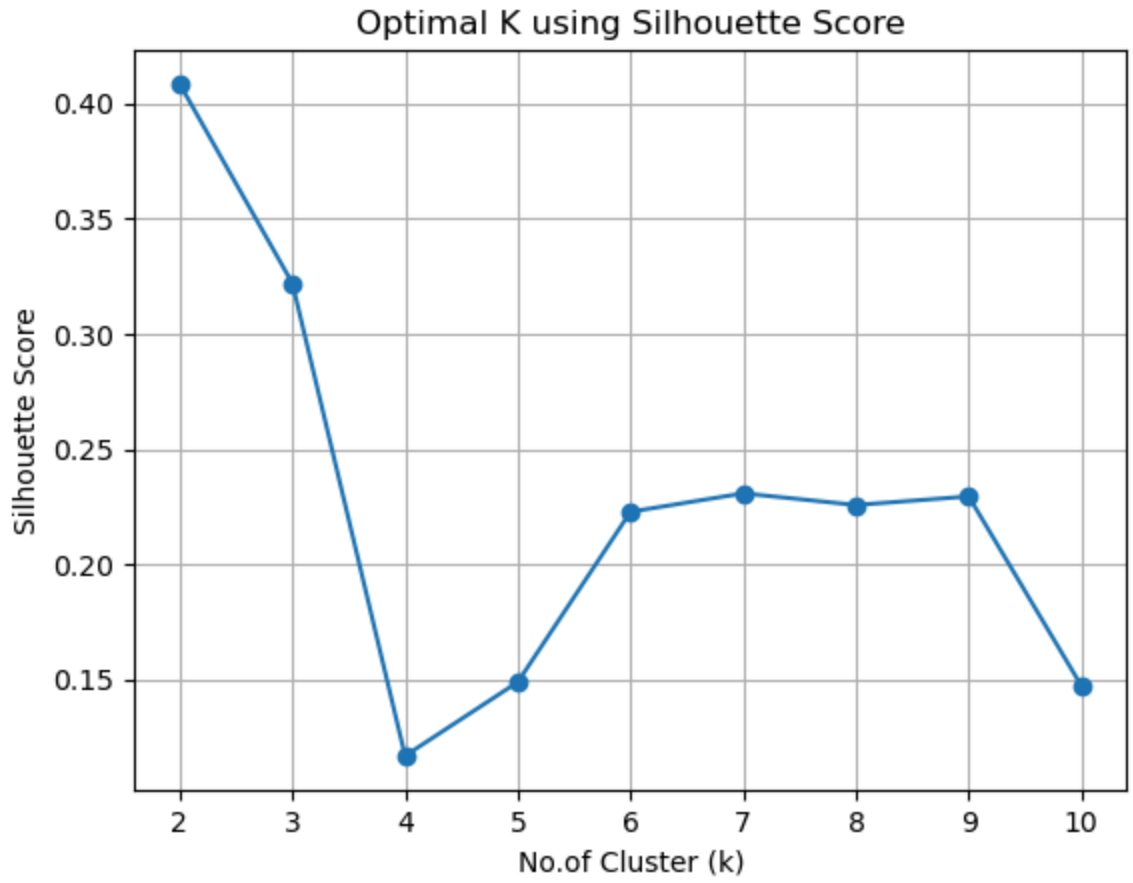
    scores.append(score)

    print(f"Cluster : {k} | Silhouette Score = {score}")
```

```
Cluster : 2 | Silhouette Score = 0.40847625085909217
Cluster : 3 | Silhouette Score = 0.3219623514333914
Cluster : 4 | Silhouette Score = 0.11687645003846443
Cluster : 5 | Silhouette Score = 0.14890043979842402
Cluster : 6 | Silhouette Score = 0.2227845328688596
Cluster : 7 | Silhouette Score = 0.23078513985838292
Cluster : 8 | Silhouette Score = 0.2257223445458049
Cluster : 9 | Silhouette Score = 0.22944093865515866
Cluster : 10 | Silhouette Score = 0.14722265293166137
```

In [16…

```
plt.plot(range(2,11), scores, marker="o")
plt.xlabel("No.of Cluster (k)")
plt.ylabel("Silhouette Score")
plt.title("Optimal K using Silhouette Score")
plt.xticks(range(2,11))
plt.grid()
plt.show()
```



In [17…
```
best_k = range(2,11)[np.argmax(scores)]
print("Optimal K :", best_k)
```

```
Optimal K : 2
```

In [20…
```
model = KMedoids(
    n_clusters=2,
    random_state=42,
    method='pam',
    metric='euclidean'
)

df['Cluster'] = model.fit_predict(x_scale)
```

In [23…
```
print("\n==== Cluster Count ====")
df['Cluster'].value_counts().sort_index()
```

Out[23]:
```
==== Cluster Count ====
 Cluster
 0    107
 1    333
 Name: count, dtype: int64
```

In [29…
```
cluster_analysis = df.groupby('Cluster')[features].mean()
```

```
print("\nAverage Spending of Each Cluster:")
cluster_analysis.round(2)
```

```
Average Spending of Each Cluster:
```

Out[29]:

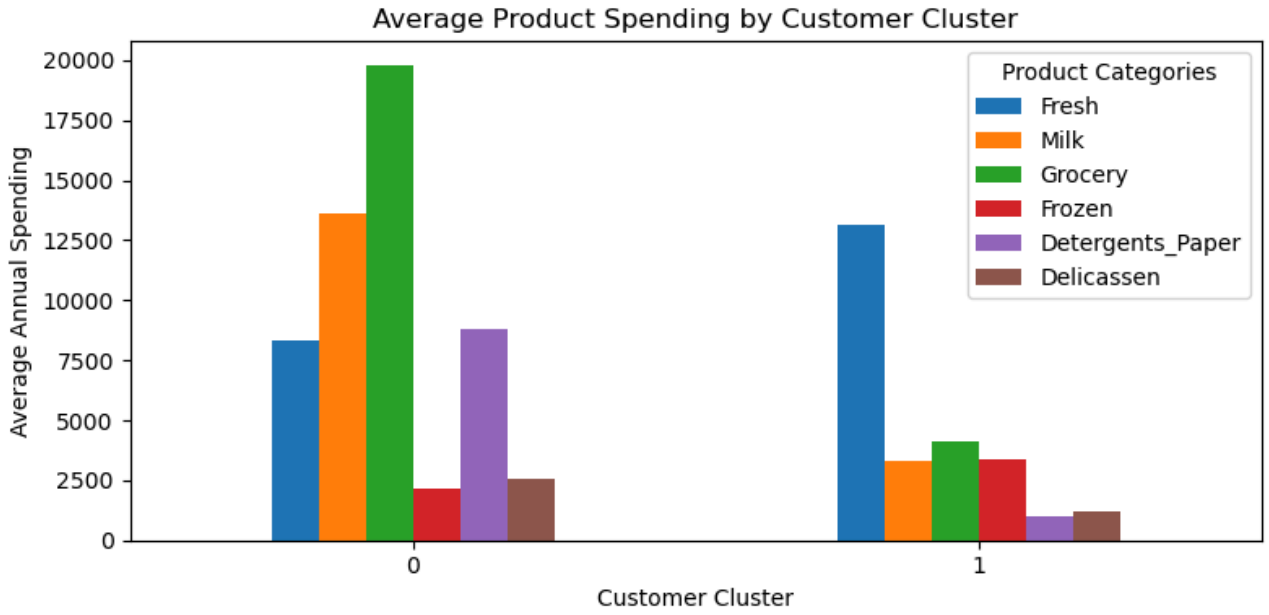
Cluster	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicassen
0	8359.74	13618.21	19798.03	2126.02	8798.78	2569.51
1	13170.09	3282.91	4144.66	3375.87	980.14	1189.20

In [33…

```
cluster_analysis.plot(
    kind='bar',
    figsize=(8, 4)
)

plt.title("Average Product Spending by Customer Cluster")
plt.xlabel("Customer Cluster")
plt.ylabel("Average Annual Spending")

plt.xticks(rotation=0)
plt.legend(title="Product Categories")
plt.tight_layout()
plt.show()
```



In [25…

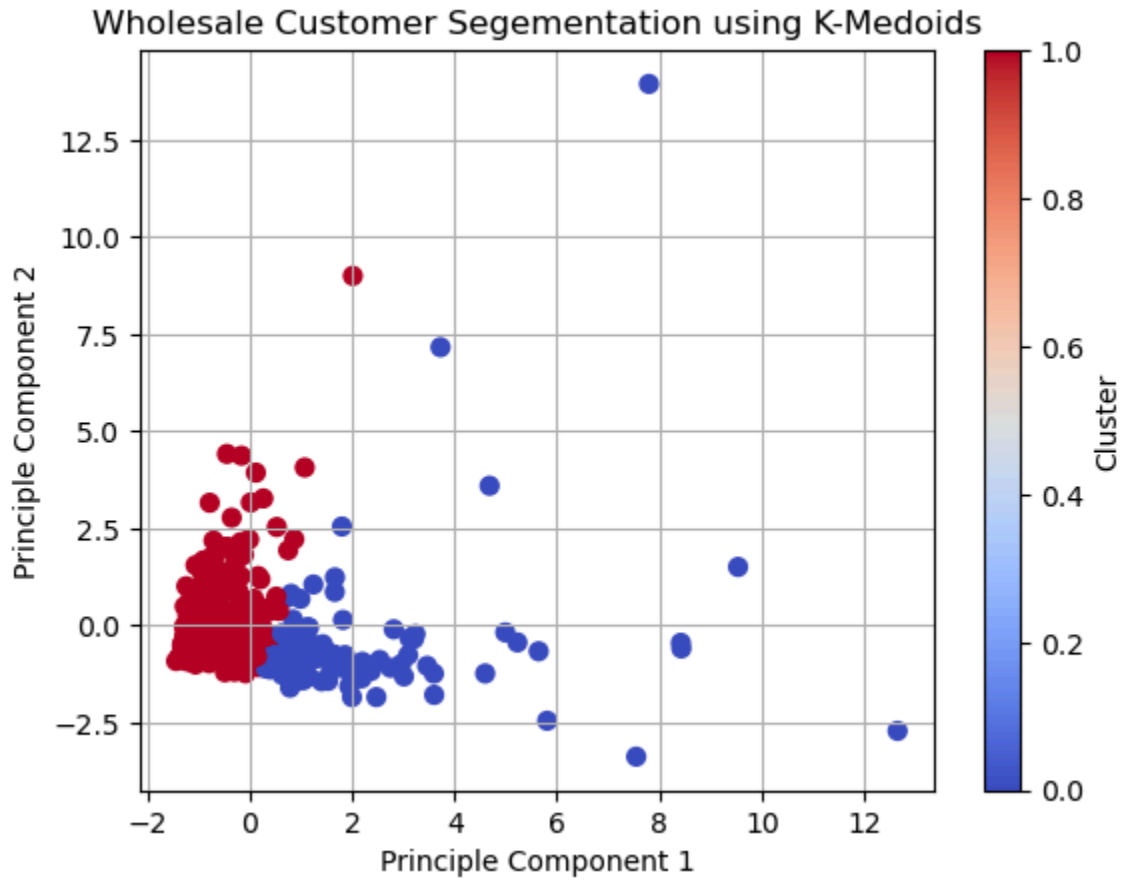
```
pca = PCA(n_components=2)
x_pca = pca.fit_transform(x_scale)
```

In [26…

```
plt.scatter(
    x_pca[:, 0],
    x_pca[:, 1],
    cmap="coolwarm",
    c=df['Cluster'],
```

```
    s=40
)

plt.xlabel("Principle Component 1")
plt.ylabel("Principle Component 2")
plt.title("Wholesale Customer Segementation using K-Medoids")
plt.colorbar(label="Cluster")
plt.grid()
plt.show()
```



In [ ]: